

TrES-2b

R-1090

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_141004 · created 2026-07-30T17:21:54+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456934.7241 +/- 0.00032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1482 +/- 0.0078
Transit depth (Rp/Rs)^2	2.2 +/- 0.23 [%]
Orbital Inclination (inc)	83.66 +/- 0.27
Airmass coefficient 1 (a1)	1.1718 +/- 0.0081
Airmass coefficient 2 (a2)	-0.1315 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	None
Optimal Aperture	3.59
Optimal Annulus	11.17
Transit Duration (day)	0.0747 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1482 $\pm$ 0.0078 vs 0.1254 (deviation 2.9 σ)
Duration fit vs published	0.0747 d vs 0.0746 d (deviation 0.03 σ)
Mid-time O-C	-0.8 min
Depth SNR	18.8
Residual scatter	0.7 %

Light curve

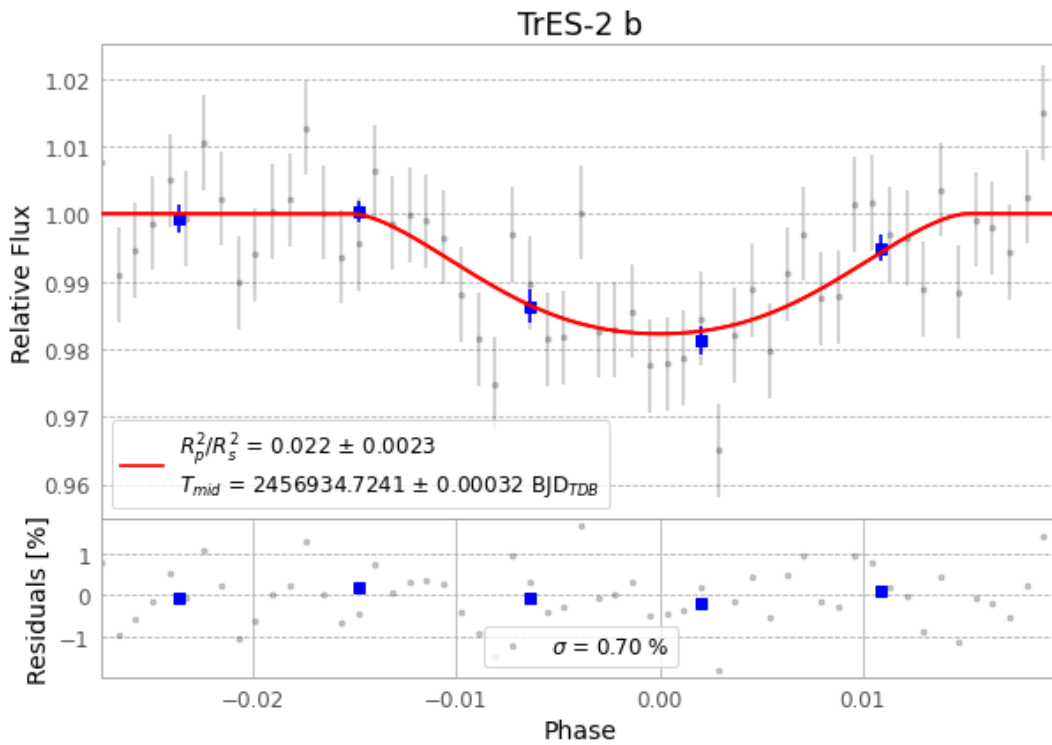


Figure 1 — FinalLightCurve_TrES-2 b_2014-10-03.png (SHA-256 588f0f5f0d84ebae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1ca8784b35488941...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

57 FITS frames (37 MB), TRES-2141004034212.FITS → TRES-2141004063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-141004.log (d1db47ba156f...), FinalLightCurve_TrES-2 b_2014-10-03.png (588f0f5f0d84...), FinalLightCurve_TrES-2 b_2014-10-03.csv (6f77d55734b0...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 17:21Z.